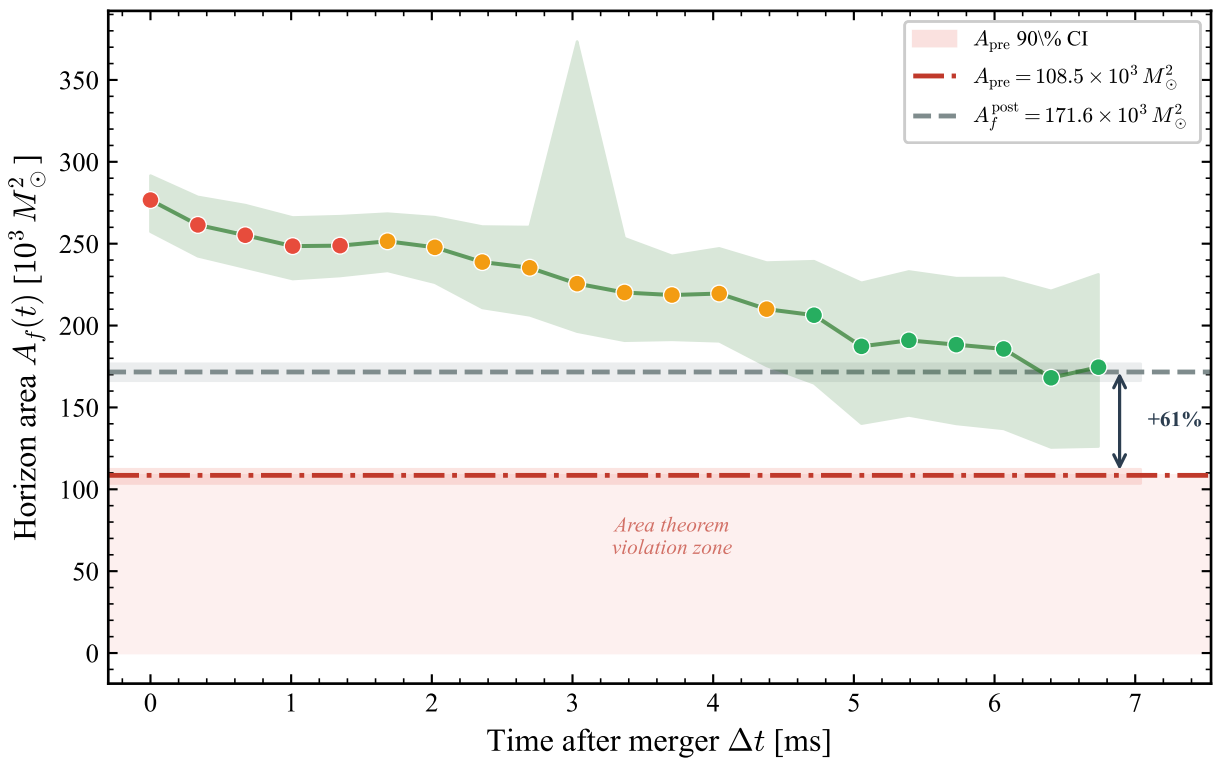


(a) Kerr Horizon Area $A_f(t)$ and Pre-Merger Bound



(b) Area Ratio $R(t) = A_f(t)/A_{\text{pre}}$

